

AZ-style identities of downsets and upsets

Tran Dan Thu

*School of Information Technology
University of Science Ho Chi Minh City
227 Nguyen Van Cu, Dist. 5
Ho Chi Minh City, Vietnam
tdt@hcmus.edu.vn*

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The Ahlswede–Zhang identity is a sharpening of the well-known LYM inequality. To establish the identity, Ahlswede and Zhang introduced an important set function called AZ function. In this paper, we present some identities which establish a relation between the AZ function and cardinalities of downsets or upsets. Some connections of these identities with the original AZ identity are considered. Furthermore, these identities yield formulas concerning the size of a star in a downset.

Keywords: LYM inequality; AZ identity; downset; upset.

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1. Introduction

Let $[n] = \{1, 2, \dots, n\}$, $2^{[n]}$ denote the family of all subsets of $[n]$, and $[n]^{(k)}$ be the family of all subsets of $[n]$ of size k . For $\mathcal{A} \subseteq 2^{[n]}$ we define the downset generated by $\mathcal{A}$ as

$$\mathcal{D}(\mathcal{A}) = \{D \subseteq [n] : D \subseteq A \text{ for some } A \in \mathcal{A}\},$$

and the upset generated by $\mathcal{A}$ as

$$\mathcal{U}(\mathcal{A}) = \{U \subseteq [n] : U \supseteq A \text{ for some } A \in \mathcal{A}\}.$$

We also write $c\mathcal{A}$ for the set of all complements of sets in $\mathcal{A}$, $c\mathcal{A} = \{[n] \setminus A : A \in \mathcal{A}\}$. If $\mathcal{A} \subseteq 2^{[n]}$ and $X \not\subseteq Y$ for all $X, Y \in \mathcal{A}$, $X \neq Y$, then $\mathcal{A}$ is called a *Sperner family* or an *antichain*.

Given a family $\mathcal{A} \subseteq 2^{[n]}$ and a subset $X \subseteq [n]$, the *Ahlsvede–Zhang (AZ) function* [1] is defined as

$$Z_{\mathcal{A}}(X) = \begin{cases} \bigcap_{X \supseteq A \in \mathcal{A}} A & \text{if } X \in \mathcal{U}(\mathcal{A}), \\ \emptyset & \text{otherwise.} \end{cases}$$

Ahlsvede and Zhang found the following powerful identity.

Theorem 1.1 ([1, Ahlsvede and Zhang]). *If $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ and $\emptyset \notin \mathcal{A}$ then*

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = 1. \quad (1.1)$$

If $\mathcal{A}$ is an antichain then $Z_{\mathcal{A}}(X) = X$ for all $X \in \mathcal{A}$. Thus for any antichain $\mathcal{A}$, the identity (1.1) implies the LYM inequality (see [5, 6, 11]) $\sum_{X \in \mathcal{A}} \frac{1}{\binom{n}{|X|}} \leq 1$.

In the next section, we present an AZ-style identity which involves the AZ function and the cardinality of an upset. Section 3 discusses intrinsic relationships between this identity and the original AZ identity. By using a duality approach as in [3, 9], we establish in Sec. 4 identities for cardinalities of downsets.

2. An Identity for Upsets

There are many generalizations or variants of the AZ identity (see [2, 4, 7, 9, 10]). The following theorem shows another variant of the AZ identity, which contains the cardinality of an upset.

Theorem 2.1. *If $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ and $\emptyset \notin \mathcal{A}$ then*

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{n+1-|Z_{\mathcal{A}}(X)|}{n+1-|X|} = 2|\mathcal{U}(\mathcal{A})| - 1. \quad (2.1)$$

We first present an induction proof of (2.1) in the style of [8]. The proof uses the following two lemmas in which the first one is an elementary identity. Another proof is given in Sec. 3, which is more illuminating and shows some connections with the original AZ identity.

Lemma 2.2. *Let b be a non-negative integer. Then $\sum_{k=0}^b \binom{b}{k} \frac{x^{k+1}}{k+1} = \frac{(1+x)^{b+1}-1}{b+1}$.*

Lemma 2.3. *Let $\mathcal{A}, \mathcal{B} \subset 2^{[n]}$, $\emptyset \notin \mathcal{A} \neq \emptyset$, $\emptyset \notin \mathcal{B} \neq \emptyset$ and put*

$$\mathcal{A} \vee \mathcal{B} = \{A \cup B : A \in \mathcal{A} \text{ and } B \in \mathcal{B}\}.$$

For any two real-valued functions $r(X)$, $s(X)$ on $\mathcal{U}(\mathcal{A} \cup \mathcal{B})$ we have

$$\begin{aligned}
 & \sum_{X \in \mathcal{U}(\mathcal{A} \cup \mathcal{B})} [r(X)|Z_{\mathcal{A} \cup \mathcal{B}}(X)| + s(X)] \\
 &= \sum_{X \in \mathcal{U}(\mathcal{A})} [r(X)|Z_{\mathcal{A}}(X)| + s(X)] + \sum_{X \in \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{B}}(X)| + s(X)] \\
 &\quad - \sum_{X \in \mathcal{U}(\mathcal{A} \vee \mathcal{B})} [r(X)|Z_{\mathcal{A} \vee \mathcal{B}}(X)| + s(X)]. \tag{2.2}
 \end{aligned}$$

Proof. If $X \in \mathcal{U}(\mathcal{A} \cup \mathcal{B})$, i.e., $X \in \mathcal{U}(\mathcal{A})$ or $X \in \mathcal{U}(\mathcal{B})$, then we can evaluate $Z_{\mathcal{A} \cup \mathcal{B}}(x)$ in three exclusive cases: $X \notin \mathcal{U}(\mathcal{A})$ and $X \in \mathcal{U}(\mathcal{B})$, $X \in \mathcal{U}(\mathcal{A})$ and $X \notin \mathcal{U}(\mathcal{B})$, $X \in \mathcal{U}(\mathcal{A})$ and $X \in \mathcal{U}(\mathcal{B})$. We can see that $Z_{\mathcal{A} \cup \mathcal{B}}(X)$ is $Z_{\mathcal{B}}(X)$ in the first case and is $Z_{\mathcal{A}}(X)$ in the second case. In the third case, by using the following identity in [3, 8]

$$|Z_{\mathcal{A} \cup \mathcal{B}}(X)| = |Z_{\mathcal{A}}(X)| + |Z_{\mathcal{B}}(X)| - |Z_{\mathcal{A} \vee \mathcal{B}}(X)|,$$

the value of $r(X)|Z_{\mathcal{A} \cup \mathcal{B}}(X)| + s(X)$ is

$$[r(X)|Z_{\mathcal{A}}(X)| + s(X)] + [r(X)|Z_{\mathcal{B}}(X)| + s(X)] - [r(X)|Z_{\mathcal{A} \vee \mathcal{B}}(X)| + s(X)].$$

Hence the left-hand side (LHS) of (2.2) is

$$\begin{aligned}
 \text{LHS}(2.2) &= \sum_{X \in \mathcal{U}(\mathcal{A}), X \notin \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{A}}(X)| + s(X)] \\
 &\quad + \sum_{X \notin \mathcal{U}(\mathcal{A}), X \in \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{B}}(X)| + s(X)] \\
 &\quad + \sum_{X \in \mathcal{U}(\mathcal{A}), X \in \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{A}}(X)| + s(X) + r(X)|Z_{\mathcal{B}}(X)| \\
 &\quad + s(X) - r(X)|Z_{\mathcal{A} \vee \mathcal{B}}(X)| - s(X)].
 \end{aligned}$$

Arranging terms of the sums we obtain

$$\begin{aligned}
 \text{LHS}(2.2) &= \sum_{X \in \mathcal{U}(\mathcal{A})} [r(X)|Z_{\mathcal{A}}(X)| + s(X)] + \sum_{X \in \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{B}}(X)| + s(X)] \\
 &\quad - \sum_{X \in \mathcal{U}(\mathcal{A}), X \in \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{A} \vee \mathcal{B}}(X)| + s(X)].
 \end{aligned}$$

Since $X \in \mathcal{U}(\mathcal{A})$ and $X \in \mathcal{U}(\mathcal{B}) \Leftrightarrow X \in \mathcal{U}(\mathcal{A} \vee \mathcal{B})$, the last expression itself is the last expression in the right-hand side (RHS) of (2.2). $\square$

Proof of Theorem 2.1. The identity (2.1) is equivalent to the following identity

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \left(2 - \frac{n+1-|Z_{\mathcal{A}}(X)|}{n+1-|X|} \right) = 1. \quad (2.3)$$

The LHS of (2.3) is

$$\text{LHS}(2.3) = \sum_{X \in \mathcal{U}(\mathcal{A})} [r(X)|Z_{\mathcal{A}}(X)| + s(X)],$$

where

$$r(X) = \frac{1}{n+1-|X|}, \quad s(X) = \frac{n+1-2|X|}{n+1-|X|}.$$

We proceed by induction on the cardinality of $\mathcal{A}$. If $|\mathcal{A}| = 1$ then $\mathcal{A} = \{A\}$ with $a = |A| > 0$. Let $b = n - a$. In this case, if $X \in \mathcal{U}(\mathcal{A})$ then $X \supseteq A$ and $Z_{\mathcal{A}}(X) = A$. Hence, we have

$$\text{LHS}(2.3) = \sum_{k=0}^{n-a} \binom{n-a}{k} \left(2 - \frac{n+1-a}{n+1-(a+k)} \right) = \sum_{k=0}^b \binom{b}{k} \left(2 - \frac{b+1}{b-k+1} \right),$$

which is

$$2 \sum_{k=0}^b \binom{b}{k} - (b+1) \sum_{k=0}^b \binom{b}{k} \frac{1}{k+1}.$$

Applying Lemma 2.2 for $x = 1$ we obtain

$$\text{LHS}(2.3) = 2 \cdot 2^b - (b+1) \frac{2^{b+1} - 1}{b+1} = 1.$$

Now we assume that the identity (2.3) holds for $1 \leq |\mathcal{A}| < h$ and consider the case $\mathcal{A} = \{A_1, A_2, \dots, A_h\}$ with $h > 1$, $\emptyset \notin \mathcal{A}$. Let $\mathcal{B} = \{A_1, A_2, \dots, A_{h-1}\}$ and $\mathcal{D} = \{A_h\}$. By using Lemma 2.3 the LHS of (2.3) is

$$\begin{aligned} \text{LHS}(2.3) &= \sum_{X \in \mathcal{U}(\mathcal{B} \cup \mathcal{D})} [r(X)|Z_{\mathcal{B} \cup \mathcal{D}}(X)| + s(X)] \\ &= \sum_{X \in \mathcal{U}(\mathcal{B})} [r(X)|Z_{\mathcal{B}}(X)| + s(X)] + \sum_{X \in \mathcal{U}(\mathcal{D})} [r(X)|Z_{\mathcal{D}}(X)| + s(X)] \\ &\quad - \sum_{X \in \mathcal{U}(\mathcal{B} \vee \mathcal{D})} [r(X)|Z_{\mathcal{B} \vee \mathcal{D}}(X)| + s(X)]. \end{aligned}$$

Since the cardinalities of $\mathcal{B}, \mathcal{D}, \mathcal{B} \vee \mathcal{D}$ are less than h , $\emptyset \notin \mathcal{B}, \emptyset \notin \mathcal{D}, \emptyset \notin \mathcal{B} \vee \mathcal{D}$, we apply the induction hypothesis to obtain $\text{LHS}(2.3) = 1 + 1 - 1 = 1$. $\square$

3. Connections with the Original AZ Identity

The identity in Theorem 2.1 is equivalent to

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|X| - |Z_{\mathcal{A}}(X)|}{n+1-|X|} = |\mathcal{U}(\mathcal{A})| - 1.$$

We investigate the quantity $|X| - |Z_{\mathcal{A}}(X)|$ by analyzing values of the AZ function. Some important properties of this quantity are found.

3.1. Some properties of AZ function

Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, $\emptyset \notin \mathcal{A}$, and $X \in \mathcal{U}(\mathcal{A})$. Notice that $Z_{\mathcal{A}}(X) \subseteq X$. The definition of the AZ function implies that $\alpha \in Z_{\mathcal{A}}(X)$ iff $X \setminus \{\alpha\} \notin \mathcal{U}(\mathcal{A})$. Hence $|Z_{\mathcal{A}}(X)|$ is the number of $\alpha \in X$ such that $X \setminus \{\alpha\} \notin \mathcal{U}(\mathcal{A})$.

For each family $\emptyset \neq \mathcal{A} \subseteq [n]^{(k)}$, we define the *lower shadow* (or *the shadow*) of $\mathcal{A}$ as

$$\Delta\mathcal{A} = \{A \subseteq [n] : |A| = k - 1, A \subset B \text{ for some } B \in \mathcal{A}\},$$

and denote $\mathcal{U}^{(k)}(\mathcal{A}) = [n]^{(k)} \cap \mathcal{U}(\mathcal{A})$. If $X \subseteq [n]$ then we simply denote $\Delta\{X\}$ by ΔX . The following two lemmas establish the relationships between the identities in this paper and the original AZ identity.

Lemma 3.1. *Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$. If $\emptyset \notin \mathcal{A}$ and $X \in \mathcal{U}(\mathcal{A})$ then*

$$|X| = |Z_{\mathcal{A}}(X)| + |\Delta X \cap \mathcal{U}(\mathcal{A})|. \quad (3.1)$$

Proof. Since $|Z_{\mathcal{A}}(X)|$ is the number of $\alpha \in X$ such that $X \setminus \{\alpha\} \notin \mathcal{U}(\mathcal{A})$, we have $|Z_{\mathcal{A}}(X)| = |\Delta X \setminus \mathcal{U}(\mathcal{A})| = |\Delta X| - |\Delta X \cap \mathcal{U}(\mathcal{A})|$. Hence the identity (3.1) holds because of $|X| = |\Delta X|$. $\square$

Lemma 3.2. *Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, $\emptyset \notin \mathcal{A}$, k be an integer, $1 \leq k \leq n$. Then*

$$\sum_{X \in \mathcal{U}^{(k)}(\mathcal{A})} [|X| - |Z_{\mathcal{A}}(X)|] = (n + 1 - k) |\mathcal{U}^{(k-1)}(\mathcal{A})|. \quad (3.2)$$

Proof. Put $r = |\mathcal{U}^{(k)}(\mathcal{A})|$, $\mathcal{U}^{(k)}(\mathcal{A}) = \{X_1, X_2, \dots, X_r\}$, $s = |\mathcal{U}^{(k-1)}(\mathcal{A})|$ and $\mathcal{U}^{(k-1)}(\mathcal{A}) = \{Y_1, Y_2, \dots, Y_s\}$. Then we define variables α_{ij} as

$$\alpha_{ij} = \begin{cases} 1 & \text{if } X_i \supset Y_j, \\ 0 & \text{otherwise.} \end{cases}$$

Since $\mathcal{U}(\mathcal{A})$ is an upset, for each $x \in [n] \setminus Y_j$ we have $Y_j \cup \{x\} \in \mathcal{U}^{(k)}(\mathcal{A})$. So the number of sets in $\mathcal{U}^{(k)}(\mathcal{A})$ which contain Y_j is $n - |Y_j| = n - (k - 1) = n + 1 - k$. Hence, $\sum_{i=1}^r \alpha_{ij} = n + 1 - k$. On the other hand, in view of Lemma 3.1, X_i contains $|X_i| - |Z_{\mathcal{A}}(X_i)|$ members of $\mathcal{U}^{(k-1)}(\mathcal{A})$. Hence, $\sum_{j=1}^s \alpha_{ij} = |X_i| - |Z_{\mathcal{A}}(X_i)|$. Finally, we evaluate the LHS of (3.2) to obtain its RHS as follows

$$\begin{aligned} \text{LHS}(3.2) &= \sum_{i=1}^r [|X_i| - |Z_{\mathcal{A}}(X_i)|] = \sum_{i=1}^r \sum_{j=1}^s \alpha_{ij} = \sum_{j=1}^s \sum_{i=1}^r \alpha_{ij} \\ &= \sum_{j=1}^s (n + 1 - k) = s(n + 1 - k) = |\mathcal{U}^{(k-1)}(\mathcal{A})|(n + 1 - k). \quad \square \end{aligned}$$

3.2. Second proof of Theorem 2.1

Since $|X| = k$ for all $X \in \mathcal{U}^{(k)}(\mathcal{A})$, we can rewrite (3.2) as

$$\sum_{X \in \mathcal{U}^{(k)}(\mathcal{A})} \frac{|X| - |Z_{\mathcal{A}}(X)|}{n + 1 - |X|} = |\mathcal{U}^{(k-1)}(\mathcal{A})|.$$

Applying the above identity for $k = 1, 2, \dots, n$ we obtain

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|X| - |Z_{\mathcal{A}}(X)|}{n + 1 - |X|} = \sum_{k=1}^n |\mathcal{U}^{(k-1)}(\mathcal{A})| = |\mathcal{U}(\mathcal{A})| - |\mathcal{U}^{(n)}(\mathcal{A})| = |\mathcal{U}(\mathcal{A})| - 1$$

which is equivalent to the identity in Theorem 2.1.

3.3. A proof of the original AZ identity

Since $|X| = k$ for each $X \in \mathcal{U}^{(k)}(\mathcal{A})$, we deduce from (3.2) for $k \geq 1$

$$\sum_{X \in \mathcal{U}^{(k)}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = \frac{|\mathcal{U}^{(k)}(\mathcal{A})|}{\binom{n}{k}} - \frac{n + 1 - k}{k \binom{n}{k}} |\mathcal{U}^{(k-1)}(\mathcal{A})|.$$

Therefore we have

$$\sum_{X \in \mathcal{U}(\mathcal{A})} \frac{|Z_{\mathcal{A}}(X)|}{|X| \binom{n}{|X|}} = \sum_{k=1}^n \left[\frac{|\mathcal{U}^{(k)}(\mathcal{A})|}{\binom{n}{k}} - \frac{|\mathcal{U}^{(k-1)}(\mathcal{A})|}{\binom{n}{k-1}} \right] = \frac{|\mathcal{U}^{(n)}(\mathcal{A})|}{\binom{n}{n}} - \frac{|\mathcal{U}^{(0)}(\mathcal{A})|}{\binom{n}{0}}$$

which is $1 - 0 = 1$ since $\emptyset \notin \mathcal{A}$.

4. Dual Identities

We can obtain the dual of identity (2.1) in the same way as in [3, 9]. The dual function of the AZ function is defined for $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$, $[n] \notin \mathcal{A}$ and $X \subseteq [n]$ as

$$Z_{\mathcal{A}}^*(X) = \begin{cases} \bigcup_{X \subseteq A \in \mathcal{A}} A & \text{if } X \in \mathcal{D}(\mathcal{A}), \\ [n] & \text{otherwise.} \end{cases}$$

For each $X \subseteq [n]$, we have the following property (see [3])

$$\overline{Z_{\mathcal{A}}^*(X)} = Z_{c\mathcal{A}}(\overline{X}), \quad (4.1)$$

in which the complement is over $[n]$, i.e., $\overline{Y} = [n] \setminus Y$ for all $Y \subseteq [n]$.

Theorem 4.1. *If $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ and $[n] \notin \mathcal{A}$ then*

$$\sum_{X \in \mathcal{D}(\mathcal{A})} \frac{1 + |Z_{\mathcal{A}}^*(X)|}{1 + |X|} = 2|\mathcal{D}(\mathcal{A})| - 1, \quad (4.2)$$

Proof. Put $\mathcal{B} = c\mathcal{A} \subseteq 2^{[n]}$, then $\mathcal{B} \neq \emptyset$ and $\emptyset \notin \mathcal{B}$. We have $X \in \mathcal{D}(\mathcal{A})$ iff $Y = \overline{X} \in \mathcal{U}(\mathcal{B})$, which implies $|\mathcal{D}(\mathcal{A})| = |\mathcal{U}(\mathcal{B})|$. By using (4.1), the LHS of (4.2) is

$$\text{LHS}(4.2) = \sum_{X \in \mathcal{D}(\mathcal{A})} \frac{1 + n - |Z_{\mathcal{B}}(\overline{X})|}{1 + n - |\overline{X}|} = \sum_{Y \in \mathcal{U}(\mathcal{B})} \frac{n + 1 - |Z_{\mathcal{B}}(Y)|}{n + 1 - |Y|}.$$

Then we use (2.1) to obtain $\text{LHS}(4.2) = 2|\mathcal{U}(\mathcal{B})| - 1 = 2|\mathcal{D}(\mathcal{A})| - 1$. $\square$

We deduce from Theorem 4.1 the following theorem for the case of stars in $2^{[n]}$.

Theorem 4.2. Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ such that $[n] \notin \mathcal{A}$. If $1 \in A$ for all $A \in \mathcal{A}$ then

$$\sum_{1 \in X \in \mathcal{D}(\mathcal{A})} \frac{|Z_{\mathcal{A}}^*(X)|}{|X|} = |\mathcal{D}(\mathcal{A})| - 1. \quad (4.3)$$

Lemma 4.3. Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ such that $[n] \notin \mathcal{A}$ and $1 \in A$ for all $A \in \mathcal{A}$. Put $\mathcal{B} = \{A \setminus \{1\} : A \in \mathcal{A}\}$ and $\mathcal{D}_1(\mathcal{A}) = \{X \in \mathcal{D}(\mathcal{A}) : 1 \in X\}$. Then

$$\mathcal{D}(\mathcal{A}) = \mathcal{D}(\mathcal{B}) \cup \mathcal{D}_1(\mathcal{A}), \quad \text{and} \quad \mathcal{D}(\mathcal{B}) \cap \mathcal{D}_1(\mathcal{A}) = \emptyset, \quad (4.4)$$

$$|\mathcal{D}_1(\mathcal{A})| = |\mathcal{D}(\mathcal{B})| = \frac{1}{2}|\mathcal{D}(\mathcal{A})|, \quad (4.5)$$

$$\text{For all } X \in \mathcal{D}_1(\mathcal{A}), \quad Y = X \setminus \{1\}, \quad |Z_{\mathcal{A}}^*(X)| = 1 + |Z_{\mathcal{B}}^*(Y)|. \quad (4.6)$$

Proof. Equation (4.4) is obvious by the definition of $\mathcal{B}$ and $\mathcal{D}_1(\mathcal{A})$. Consider the mapping $X \mapsto X \cup \{1\}$ from $\mathcal{D}(\mathcal{B})$ to $\mathcal{D}_1(\mathcal{A})$. This mapping is injective, so $|\mathcal{D}(\mathcal{B})| \leq |\mathcal{D}_1(\mathcal{A})|$. Similarly, the injective mapping $X \mapsto X \setminus \{1\}$ from $\mathcal{D}_1(\mathcal{A})$ to $\mathcal{D}(\mathcal{B})$ gives $|\mathcal{D}_1(\mathcal{A})| \leq |\mathcal{D}(\mathcal{B})|$. Hence $|\mathcal{D}(\mathcal{B})| = |\mathcal{D}_1(\mathcal{A})|$ which implies the first equality in (4.5). The second one follows from (4.4). For all $X \in \mathcal{D}_1(\mathcal{A})$ and $Y = X \setminus \{1\}$, we have

$$\begin{aligned} Z_{\mathcal{A}}^*(X) &= \bigcup_{X \subseteq A \in \mathcal{A}} A = \left(\bigcup_{X \subseteq A \in \mathcal{A}} (A \setminus \{1\}) \right) \cup \{1\} \\ &= \left(\bigcup_{Y \subseteq B \in \mathcal{B}} B \right) \cup \{1\} = Z_{\mathcal{B}}^*(Y) \cup \{1\} \end{aligned}$$

which implies (4.6) because of $1 \notin Z_{\mathcal{B}}^*(Y)$. $\square$

Proof of Theorem 4.2. Let $\mathcal{A}$, $\mathcal{B}$, and $\mathcal{D}_1(\mathcal{A})$ be the same as in Lemma 4.3. The LHS of (4.3) is

$$\text{LHS}(4.3) = \sum_{Y \in \mathcal{D}(\mathcal{B})} \frac{|Z_{\mathcal{A}}^*(Y \cup \{1\})|}{|Y \cup \{1\}|} = \sum_{Y \in \mathcal{D}(\mathcal{B})} \frac{1 + |Z_{\mathcal{B}}^*(Y)|}{1 + |Y|}$$

which is $2 \cdot |\mathcal{D}(\mathcal{B})| - 1 = |\mathcal{D}(\mathcal{A})| - 1$. $\square$

In fact, the quantity $\frac{1}{2}|\mathcal{D}(\mathcal{A})|$ is the size of the star $\mathcal{S}(1, \mathcal{A}) = \{X \in \mathcal{D}(\mathcal{A}) : 1 \in X\}$ in the downset $\mathcal{D}(\mathcal{A})$ with the condition $1 \in A$ for all $A \in \mathcal{A}$. The following corollary shows an identity of stars in any downset $\mathcal{D}(\mathcal{A})$ without this condition.

Corollary 4.4. *Let $\emptyset \neq \mathcal{A} \subseteq 2^{[n]}$ such that $[n] \notin \mathcal{A}$. For each $j \in \bigcup_{A \in \mathcal{A}} A$, we consider $\mathcal{S}(j, \mathcal{A}) = \{X \in \mathcal{D}(\mathcal{A}) : j \in X\}$. Then*

$$\sum_{X \in \mathcal{S}(j, \mathcal{A})} \frac{|Z_{\mathcal{A}}^*(X)|}{|X|} = 2|\mathcal{S}(j, \mathcal{A})| - 1. \quad (4.7)$$

Proof. For each $j \in \bigcup_{A \in \mathcal{A}} A$, we put $\mathcal{A}_j = \{A \in \mathcal{A} : j \in A\} \neq \emptyset$, then $\mathcal{S}(j, \mathcal{A}_j) = \mathcal{S}(j, \mathcal{A})$. By Lemma 4.3 we have $|\mathcal{D}(\mathcal{A}_j)| = 2 \cdot |\mathcal{S}(j, \mathcal{A}_j)| = 2 \cdot |\mathcal{S}(j, \mathcal{A})|$. If $X \in \mathcal{S}(j, \mathcal{A})$ then $Z_{\mathcal{A}}^*(X) = Z_{\mathcal{A}_j}^*(X)$. Finally, we apply Theorem 4.2 to obtain

$$\sum_{j \in X \in \mathcal{D}(\mathcal{A}_j)} \frac{|Z_{\mathcal{A}_j}^*(X)|}{|X|} = |\mathcal{D}(\mathcal{A}_j)| - 1$$

which is equivalent to (4.7). □

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